

Q & Soergel 9: Soergel theory 1

1) Introduction

2) Properties of $P(\lambda_-)$

1) Introduction

In Sec 2 of Lec 1 we mentioned the Soergel functor $V := \text{Hom}_{\mathcal{O}^{\text{pr}}} (P(w_0 \cdot 0), ?) : \mathcal{O}^{\text{pr}} \rightarrow \text{End}_{\mathcal{O}^{\text{pr}}} (P(w_0 \cdot 0))^{\text{opp-mod}}$ and stated three theorems describing its properties including Thm 2: that it's fully faithful on projectives. Our next goal is to prove a more general version of this claim.

We will work over a char. 0 field $\mathbb{K}$ and take $\mathfrak{g}=0$. We skip the subscript 0: $\mathcal{O} := \mathcal{O}_0$, $\Delta(\lambda) := \Delta_0(\lambda)$, etc. Let $\tilde{\Sigma} = W \cdot \lambda \in \Lambda / (W \cdot \cdot)$. We'll write λ^+ (resp., λ^-) for the unique element of $\tilde{\Sigma}$ s.t. $\lambda^{\pm} + \rho$ is dominant (resp., anti-dominant). In particular, consider the unique projective $P(\lambda_-) \in \mathcal{O}^{\tilde{\Sigma}}$ (see Sec 2.3 of Lec 8) so that

$$\dim \text{Hom}_{\mathcal{O}} (P(\lambda_-), L(\mu)) = \delta_{\lambda^-, \mu}$$

Set $E (= E_{\tilde{\Sigma}}) := \text{End}_{\mathcal{O}}(P(\lambda^-))^{\text{opp}}$ - we'll compute this algebra later. We get the exact functor

$$V_{\lambda} := \text{Hom}_{\mathcal{O}}(P(\lambda^-), ?) : \mathcal{O}^{\tilde{\Sigma}} \longrightarrow E\text{-mod}_{f.g.}$$

Our main result that we'll start proving in this lecture is:

Thm: Let $M, P \in \mathcal{O}^{\tilde{\Sigma}}$ & P be projective. Then the map $\text{Hom}_{\mathcal{O}}(M, P) \xrightarrow{\sim} \text{Hom}_E(V_{\lambda}(M), V_{\lambda}(P))$ induced by V_{λ} is iso. In particular, V_{λ} is fully faithful on projectives.

Let us explain ideas of the proof. Note that

- i) V_{λ} admits a right adjoint, to be denoted by $V_{\lambda}^!$
- ii) Thm $\Leftrightarrow$ adjunction unit $\eta_P : P \rightarrow V_{\lambda}^! V_{\lambda} P$ is iso.

To see i), recall that $\mathcal{O}^{\tilde{\Sigma}} \xrightarrow{\sim} \tilde{E}\text{-mod}_{f.g.}$, where $\tilde{E} = \text{End}_{\mathcal{O}}(\tilde{P})^{\text{opp}}$ for a projective generator $\tilde{P}$ of $\mathcal{O}^{\tilde{\Sigma}}$ (Sec 2.1 of Lec 8). $\tilde{E}$ is fin. dim. hence Noetherian. Now

2] we use the following:

Observation: Let $\tilde{E}, E$ be Noetherian associative unital rings & $\mathcal{F}: \tilde{E}\text{-mod}_{fg} \rightarrow E\text{-mod}_{fg}$ be a right exact functor. Then $\mathcal{F}(\tilde{E})$ is an $E\text{-}\tilde{E}$ -bimodule, $\mathcal{F} \simeq \mathcal{F}(\tilde{E}) \otimes_{\tilde{E}} ?$, and a right adjoint of $\mathcal{F}$ is given by $\text{Hom}_E(\mathcal{F}(\tilde{E}), ?)$

(ii) is an **exercise** on basic abstract nonsense.

Now note that for $M_1, M_2 \in \mathcal{O}^{\tilde{\Sigma}}$, $\eta_{M_1 \oplus M_2} = \eta_{M_1} \oplus \eta_{M_2}$. So $\eta_{M_1 \oplus M_2}$ is an isomorphism $\Leftrightarrow$ both η_{M_1}, η_{M_2} are iso's. Recall the projective generator $\tilde{P} = \mathcal{I}_{\lambda' \rightarrow \lambda^+} \left(\bigoplus_{w \in W} \binom{+}{w} \right) \Delta(\lambda')$, where λ' is dominant & $\lambda' - \lambda^+$ is dominant. By Sec 2.3 in Lec 8

$$\tilde{P} \simeq \bigoplus_{\lambda \in \tilde{\Sigma}} P(\lambda)^{\oplus m_\lambda}, \quad P \simeq \bigoplus_{\lambda \in \Sigma} P(\lambda)^{\oplus n_\lambda} \quad (m_\lambda \geq 0, n_\lambda \geq 0 \forall \lambda)$$

So P is a direct summand in $\tilde{P}^{\oplus k}$ for some k & Thm follows from

Proposition: $\eta_{\tilde{P}}: \tilde{P} \xrightarrow{\sim} V_{\lambda'}^! V_{\lambda} \tilde{P}$.

This will be deduced from the following 2 claims

Important claim 1: $P(\lambda^-)$ is iso

Important claim 2: $\exists$ exact sequence
 $0 \rightarrow \tilde{P} \rightarrow P(\lambda^-)^{\oplus m} \rightarrow P(\lambda^-)^{\oplus n}$ for some $m, n > 0$

Important exercise: Deduce Proposition from these two claims (hint: $V_\lambda^!$ is left exact; use 5-lemma).

Proofs of Important Claims is based on

Important claim 3: $P(\lambda^-) \xrightarrow{\sim} \mathcal{T}_{-\rho \rightarrow \lambda^+} \Delta(-\rho)$.

2) Properties of $P(\lambda^-)$

2.1) Anti-dominant Verma $\Delta(\lambda^-)$

Thm: 1) $\Delta(\lambda^-)$ is simple.

2) $\forall \lambda \in \Sigma, \Delta(\lambda^-)$ is the only simple sub of $\Delta(\lambda)$
& not a composition factor of $\Delta(\lambda)/\Delta(\lambda^-)$.

Proof:

1): In Sec 1 of Lec 6, for $M \in \mathcal{O}^{\Xi}$ we introduced

$$d^{\Xi}(M) := \sum_{\mu \in \Xi} \dim M_{\mu} \in \mathbb{Z}_{\geq 0}$$

which bounds the length of M from above. Since λ^{-} is minimal in Ξ & $\Delta(\lambda^{-})_{\mu} \neq 0 \Rightarrow \mu \leq \lambda^{-}$, we get

$$d^{\Xi}(\Delta(\lambda^{-})) = \dim \Delta(\lambda^{-})_{\lambda^{-}} = 1$$

hence $\Delta(\lambda^{-})$ is simple.

2) We'll first show that $\Delta(\lambda)$ contains a unique irreducible subobject that has to be isomorphic to $\Delta(\lambda^{-})$. We'll then apply a translation functor to show that $\Delta(\lambda^{-})$ doesn't appear as a subquotient of $\Delta(\lambda)/\Delta(\lambda^{-})$.

Step 1: We know (Sec 1 of Lec 6) that $\Delta(\lambda)$ has finite length, hence a simple subobject, say L .

Step 2: Note that $\mathcal{U}(\kappa^{-})$ has no 0-divisors (exercise:

5 | use that $\text{gr } \mathcal{U}(\kappa^{-}) \simeq S(\kappa^{-})$, a polynomial algebra).

Since $\Delta(\lambda)$ is a free (rk 1) $\mathcal{U}(\mathfrak{h}^-)$ -module, $\mathcal{L}$ is torsion-free over $\mathcal{U}(\mathfrak{h}^-)$.

Step 3: Let $\mathcal{L}' = \mathcal{L}(\lambda')$ for λ' . If $\lambda' \neq \lambda^-$, then $\exists i$ s.t. $\kappa := \langle \lambda'_i + \rho, \alpha_i^\vee \rangle > 0$. Hence $f_i^\kappa v_\lambda \in M_{s_i \cdot \lambda}^{\mathfrak{h}} \setminus \{0\} \xrightarrow{\sim} \Delta(s_i \cdot \lambda) \xrightarrow{\varphi} \Delta(\lambda')$ w. $f_i^\kappa v_\lambda \in \text{im } \varphi \Rightarrow f_i^\kappa v_\lambda = 0$ in $\mathcal{L}(\lambda')$ $\leadsto$ contradiction w. Step 2 $\Rightarrow \lambda' = \lambda^-$.

Step 4: By part 1), $\mathcal{L}(\lambda^-) = \Delta(\lambda^-)$. The translation functor $\mathcal{T}_{\lambda^+ \rightarrow -\rho}$ is exact & $\mathcal{T}_{\lambda^+ \rightarrow -\rho} \Delta(\lambda) = \Delta(-\rho) \neq 0 \ \forall \lambda \in \Sigma$ (Thm in Sec 1 of Lec 7). So $\mathcal{T}_{\lambda^+ \rightarrow -\rho}(\Delta(\lambda)/\Delta(\lambda^-)) = 0 \Rightarrow \Delta(\lambda^-)$ is not a composition factor in $\Delta(\lambda)/\Delta(\lambda^-)$. $\square$

Exercise: Prove that $\forall \lambda, \lambda' \in \Sigma \Rightarrow \dim \text{Hom}_0(\Delta(\lambda), \Delta(\lambda')) \leq 1$ & any nonzero homomorphism $\Delta(\lambda) \rightarrow \Delta(\lambda')$ is injective.

2.2) $P(\lambda^-) \xrightarrow{\sim} \mathcal{T}_{-\rho \rightarrow \lambda^+} \Delta(-\rho)$ (Important Claim 3)

This claim follows from

Proposition: 1) $P := \mathcal{T}_{-p \rightarrow \lambda^+} \Delta(-p)$ is projective & satisfies

$$\dim \operatorname{Hom}_{\mathcal{O}}(P, \Delta(\lambda)) = 1 \quad \forall \lambda \in \Sigma$$

2) Let $r \geq 0$ & P be a projective in $\mathcal{O}^{\Sigma}$. Suppose

$$\dim \operatorname{Hom}_{\mathcal{O}}(P, \Delta(\lambda)) = r \quad \forall \lambda \in \Sigma \quad (1)$$

Then $P \simeq P(\lambda^-)^{\oplus r}$.

Proof: 1) $\Delta(-p)$ is projective in $\mathcal{O}^{l-p}$ & $\mathcal{T}_{-p \rightarrow \lambda^+}$ sends projectives to projectives so $\mathcal{T}_{-p \rightarrow \lambda^+} \Delta(-p)$ is projective. Also

$$\operatorname{Hom}_{\mathcal{O}^{\Sigma}}(\mathcal{T}_{-p \rightarrow \lambda^+} \Delta(-p), \Delta(\lambda)) \xrightarrow{\sim} \operatorname{Hom}_{\mathcal{O}^{-p}}(\Delta(-p), \mathcal{T}_{\lambda^+ \rightarrow -p} \Delta(\lambda)) \xrightarrow{\sim}$$

[1) of Thm in Sec 1 of Lec 7] $\operatorname{End}_{\mathcal{O}^{-p}}(\Delta(-p)) \xrightarrow{\sim} \mathbb{K}$.

2) Let now P be a projective satisfying (1). Write $\mathcal{F}$ for $\operatorname{Hom}_{\mathcal{O}}(P, ?): \mathcal{O}^{\Sigma} \rightarrow \operatorname{Vect}$, it's exact. Thanks to Thm in Sec 2.1, $\mathcal{L}(\lambda^-) = \Delta(\lambda^-)$. So $\mathcal{F}(\mathcal{L}(\lambda^-)) \simeq \mathcal{F}(\Delta(\lambda^-)) \simeq \mathbb{K}^r$. Also,

$$\mathcal{F}[\Delta(\lambda)/\Delta(\lambda^-)] \xrightarrow{\sim} \mathcal{F}(\Delta(\lambda))/\mathcal{F}(\Delta(\lambda^-)) = \mathbb{K}^r/\mathbb{K}^r = 0. \text{ But for}$$

$\lambda \neq \lambda^-$, we have $\Delta(\lambda)/\Delta(\lambda^-) \rightarrow \mathcal{L}(\lambda)$ so $\mathcal{F}(\mathcal{L}(\lambda)) = 0$. By Sec

2.3 in Lec 8, $P \simeq P(\lambda^-)^{\oplus r}$. $\square$

2.3) $\varphi_{P(\lambda^-)}: P(\lambda^-) \xrightarrow{\sim} \mathbb{V}_\lambda^! \mathbb{V}_\lambda P(\lambda^-)$ (Important Claim 1)

Lemma: 1) $\forall M \in \mathcal{O}^\Sigma, N \subset \mathbb{V}_\lambda^! \mathbb{V}_\lambda M$. Then $N \neq 0 \Rightarrow \mathbb{V}_\lambda(N) \neq 0$.

2) $\forall M \in \mathcal{O}^\Sigma \Rightarrow \mathbb{V}_\lambda(\ker \varphi_M) = \mathbb{V}_\lambda(\operatorname{coker} \varphi_M) = 0$

3) If M is filtered by Vermas, then φ_M is injective

Sketch of proof:

1) & 2) are general properties of adjoint functors between abelian categories and are left as **exercises**.

3) reduces to the case when $M = \Delta(\lambda)$ w. $\lambda \in \Sigma$. Any nonzero subobject of $\Delta(\lambda)$ contains $\Delta(\lambda^-)$ by Thm in Sec 2.1 & $\mathbb{V}_\lambda(\Delta(\lambda^-)) \neq 0$. Now 3) follows from 2). $\square$

Proof of Important Claim 1:

Step 1: $P(\lambda^-) \simeq \mathcal{T}_{-p \rightarrow \lambda^+} \Delta(-p)$ by Important Claim 3.

By Thm in Sec 1 of Lec 7, the r.h.s. is filtered by Vermas, so by 3) of Lemma, $\varphi_{P(\lambda^-)}$ is injective. Write N for $\operatorname{coker} \varphi_{P(\lambda^-)}$.

8 | Step 2: We claim that $\mathcal{T}_{-p \rightarrow \lambda^+} \Delta(-p)$ is injective.

Indeed $\mathcal{O}^p$ is a semisimple category ($\text{Vect}_{\mathbb{F}_d}$) so every object including $\Delta(-p)$ is injective. And $\mathcal{T}_{-p \rightarrow \lambda^+}$ admits a bi-adjoint so sends injectives to injectives yielding the claim.

Step 3: By Step 2, $0 \rightarrow P(-\lambda) \rightarrow \mathbb{V}_{\lambda}^! \mathbb{V}_{\lambda} P(-\lambda) \rightarrow N \rightarrow 0$ splits so N is a sub in $\mathbb{V}_{\lambda}^! \mathbb{V}_{\lambda} P(-\lambda)$. By 2) of Lemma, $\mathbb{V}_{\lambda}(N) = 0$. By 1) of Lemma, $N = 0$, finishing the proof. $\square$

Important remark: Similarly to 3) of Lemma we see that the adjunction unit $\eta_M': M \rightarrow \mathcal{T}_{-p \rightarrow \lambda^+} \mathcal{T}_{\lambda^+ \rightarrow -p} M$ is injective if M is filtered by Vermas.

2.4) Toward Important Claim 2

Lemma. $\forall$ dominant $\lambda' \exists$ an exact sequence

$$0 \rightarrow \Delta(\lambda') \rightarrow P(\lambda'^-) \rightarrow P(\lambda'^-)^{\oplus n}$$

Proof: $P(\lambda'^-) \simeq \mathcal{T}_{-p \rightarrow \lambda'} \Delta(-p)$ admits a filtration by Vermas w. $\Delta(\lambda')$ (the largest label in the block) appearing as a sub (Thm in Sec 1 of Lec 7).

This gives an embedding $\Delta(\lambda') \hookrightarrow P(\lambda')^-$ whose cokernel, M , is filtered by Verma's. By Important remark in Sec 2.3 we have injective adjunction unit $\eta'_M: M \rightarrow \mathcal{T}_{-\rho \rightarrow \lambda'} \mathcal{T}_{\lambda' \rightarrow -\rho} M \simeq [\mathcal{T}_{\lambda' \rightarrow -\rho} M \simeq \Delta(-\rho)^{\oplus n}$, $n = |W| - 1]$ $(\mathcal{T}_{-\rho \rightarrow \lambda'} \Delta(-\rho))^{\oplus n} \simeq P(\lambda')^{-\oplus n}$ $\square$